

Technical Appendix: Multi-Trait Sample Overlap

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This appendix develops the cross-trait phenotype and sample-overlap information required by multi-trait Bayesian linear regression (MTBLR) from summary statistics. It preserves the **SSY** and overlap derivations from the archived theory draft, connects them to the current **sblr** multi-trait interfaces, and distinguishes mathematically required quantities from current implementation constraints.

The central issue is simple: separate single-trait summary statistics preserve each trait’s marginal phenotype sum of squares, but they do not preserve cross-trait phenotype products. A joint model with residual covariance needs those cross-products or defensible approximations to them.

1 Multi-Trait Individual-Level Model

Let T denote the number of traits. For a fully overlapping sample, define:

- Y , an $n \times T$ phenotype matrix;
- X , an $n \times m$ genotype matrix;
- B , an $m \times T$ marker-effect matrix; and
- E_Y , an $n \times T$ residual matrix.

The multi-trait linear model is

$$Y = XB + E_Y.$$

For individual i , the residual vector may be modeled as

$$e_i \sim N_T(0, \Sigma_e),$$

where Σ_e is the residual covariance matrix. Marker-effect vectors may also share covariance. For example, an active marker-effect vector b_j may have a prior involving a covariance matrix Σ_b .

These two covariance structures have different meanings:

- Σ_b describes prior or posterior covariance among marker effects across traits and enables genetic information sharing;
- Σ_e describes covariance among residual phenotype deviations for individuals observed on the same traits.

Residual covariance is therefore directly tied to which individuals are shared across traits.

2 Why Marginal Trait Statistics Are Not Enough

For one trait t , the phenotype-side scalar summary is

$$yy_t = y_t^\top y_t = \sum_i y_{it}^2.$$

Together with $X^\top y_t$, genotype cross-products, and n_t , this can support a single-trait likelihood. Repeating those inputs for several traits supports separate or parallel single-trait fits.

A joint multi-trait likelihood additionally depends on cross-trait phenotype products:

$$y_t^\top y_s = \sum_i y_{it} y_{is}.$$

The marginal values yy_t and yy_s do not determine this cross-product. For example, two phenotype pairs can have identical marginal sums of squares but opposite cross-trait correlations.

Consequently, a full residual covariance update cannot be reconstructed from trait-specific yy values alone. It requires the complete cross-trait phenotype cross-product matrix, usually denoted SSY or $Y^\top Y$, together with a clear sample-overlap interpretation.

3 Multi-Trait Summary Statistics

Under full sample overlap with one common X , the central summary statistics are:

$$WY = X^\top Y, \quad XX = X^\top X, \quad SSY = Y^\top Y, \quad n.$$

Their dimensions are:

Quantity	Dimension	Meaning
$WY = X^\top Y$	$m \times T$	Marker-trait cross-products
$XX = X^\top X$	$m \times m$	Marker cross-products or compatible LD representation
$SSY = Y^\top Y$	$T \times T$	Trait cross-products
n	scalar	Number of jointly observed individuals

For partially overlapping studies, the representation expands to include:

- trait-specific sample sizes n_t ;
- pairwise overlap counts N_{ts} ; and
- trait-pair-specific interpretations of cross-products.

Define the sample-overlap matrix

$$N = \begin{bmatrix} N_{11} & \cdots & N_{1T} \\ \vdots & \ddots & \vdots \\ N_{T1} & \cdots & N_{TT} \end{bmatrix},$$

where

$$N_{ts} = |\mathcal{J}_{ts}|$$

and $\mathcal{J}_{ts}$ is the set of individuals observed for both traits t and s . By definition,

$$N_{tt} = n_t.$$

4 Cross-Trait SSY

For partially overlapping traits, define

$$SSY_{ts} = \sum_{i \in \mathcal{J}_{ts}} y_{it} y_{is}.$$

The diagonal entries reduce to ordinary single-trait sums of squares:

$$SSY_{tt} = yy_t.$$

When all traits are observed on exactly the same individuals, $\mathcal{J}_{ts}$ is the common sample for every pair and

$$SSY = Y^\top Y$$

has its ordinary matrix-cross-product meaning.

When samples overlap only partially, **SSY** remains a useful symmetric matrix of pairwise cross-products, but its entries are calculated from different sets of individuals. It should not be interpreted as the cross-product of one complete rectangular phenotype matrix unless such a matrix actually exists.

5 Full Sample Overlap

The cleanest MTBLR setting has all traits measured on the same n individuals:

$$n_t = n, \quad N_{ts} = n \quad \text{for all } t, s.$$

Then:

$$WY = X^\top Y, \quad SSY = Y^\top Y$$

can be computed directly, and the residual covariance has the conventional interpretation:

$$\Sigma_e = \text{Cov}(e_i)$$

for residual vectors observed jointly on the same individuals.

The current CSR MTBLR kernel is closest to this setting. It uses shared LD, requires equal trait sample sizes and matching `ww` values across traits, and accepts a full trait cross-product matrix through its `yy` argument.

The archived theory draft also defines a weaker “full-overlap” approximation for unequal marginal sample sizes:

$$N_{ts}^{\text{full}} = \min(n_t, n_s).$$

This assumes the smaller study is nested within the larger study for each trait pair. It is useful as an explicit approximation, but it is not equivalent to all traits being measured on one common sample.

6 Partial Sample Overlap

With partial overlap, different trait pairs may share different individuals:

$$0 < N_{ts} < \min(n_t, n_s).$$

A convenient parameterization uses overlap fractions $p_{ts} \in [0, 1]$:

$$N_{ts}^{\text{frac}} = p_{ts} \min(n_t, n_s), \quad N_{tt} = n_t.$$

The matrix N records the amount of direct information available about each trait pair. A cross-trait phenotype covariance estimated from 500 shared individuals is not supported by the same information as one estimated from 50,000 shared individuals, even when the numerical correlation is identical.

Partial overlap complicates the likelihood because:

- $\mathbf{SSY}$ entries arise from different pair-specific samples;
- marker-trait statistics $X_t^\top y_t$ may arise from different genotype samples;
- LD and diagonal genotype cross-products may differ by trait;
- residual covariance reflects only shared individuals; and
- a single common $X^\top X$ may not exactly represent every trait pair.

These issues require explicit modeling assumptions. Supplying an overlap matrix does not by itself make a shared-LD likelihood exact.

7 No Overlap and Unknown Overlap

If traits t and s have no shared individuals,

$$N_{ts} = 0.$$

There is then no direct individual-level cross-product $\sum_i y_{it}y_{is}$ for that pair. Setting

$$\mathbf{SSY}_{ts} = 0$$

records the absence of observed cross-trait phenotype product under a centered phenotype convention. It does not establish that biological, genetic, or population-level residual correlation is zero.

Genetic covariance can still be informed by shared markers, aligned marker-trait associations, and an appropriate cross-study model. Residual covariance between non-overlapping samples, however, is not directly identified by observed paired residuals. It must be fixed, strongly regularized, inferred from external information, or excluded from the model.

When overlap is unknown, off-diagonal $\mathbf{SSY}$ is also unknown. Treating unknown overlap as full overlap can create spurious residual covariance; treating it as zero can discard real correlated-error information. Sensitivity analyses over plausible overlap assumptions are preferable to silently choosing either extreme.

8 Phenotypic Variance From Marginal Summary Statistics

For trait t , let:

$$yy_t = \sum_{i \in \mathcal{I}_t} y_{it}^2, \quad n_t = |\mathcal{I}_t|.$$

If phenotypes are centered or approximately centered, an empirical variance estimate is

$$\widehat{\text{Var}}(y_t) \approx \frac{yy_t}{n_t}.$$

Using $n_t - 1$ instead gives the usual sample-variance denominator. The choice must match the intended likelihood and preprocessing convention. The archived reconstruction uses yy_t/n_t as the cross-product-per-observation quantity.

Collect the marginal variance estimates in:

$$\hat{\sigma}_y^2 = \left(\frac{yy_1}{n_1}, \dots, \frac{yy_T}{n_T} \right).$$

If phenotypes are not centered, yy_t/n_t is a second raw moment rather than a variance. Covariance reconstruction then requires explicit treatment of trait means.

9 Covariance Reconstruction From Correlations

Suppose a phenotypic correlation matrix

$$R_y = [r_{ts}]$$

is available from individual-level data, a reported study estimate, or an external source. Convert it to phenotype covariance units using:

$$\hat{\sigma}_{ts} = r_{ts} \sqrt{\widehat{\text{Var}}(y_t) \widehat{\text{Var}}(y_s)}.$$

In matrix form:

$$\widehat{\text{Cov}}(Y) = D_y R_y D_y,$$

where

$$D_y = \text{diag} \left(\sqrt{\widehat{\text{Var}}(y_1)}, \dots, \sqrt{\widehat{\text{Var}}(y_T)} \right).$$

This rescaling is essential. A correlation matrix alone is dimensionless and cannot replace phenotype cross-products when traits have different scales.

The supplied R_y must have the same trait order as $\mathbf{yy}$, n , $\mathbf{WY}$, and all covariance priors. If R_y comes from another sample, its transportability is an additional assumption.

10 Reconstructing SSY

Under the approximation that pairwise covariance is homogeneous across individuals and that missingness or study membership does not alter it,

$$E(y_{it}y_{is}) = \sigma_{ts}$$

for centered traits observed together. Therefore:

$$\begin{aligned} E(SSY_{ts}) &= E \left[\sum_{i \in \mathcal{I}_{ts}} y_{it}y_{is} \right] \\ &= \sum_{i \in \mathcal{I}_{ts}} E(y_{it}y_{is}) \\ &= N_{ts}\sigma_{ts}. \end{aligned}$$

Replacing σ_{ts} with its estimate gives:

$$\widehat{SSY}_{ts} = N_{ts}\hat{\sigma}_{ts}.$$

In matrix notation:

$$\widehat{SSY} = N \odot \widehat{\text{Cov}}(Y),$$

where $\odot$ denotes elementwise multiplication.

The diagonal should be forced to the observed marginal sums of squares:

$$\widehat{SSY}_{tt} = yy_t.$$

This avoids small inconsistencies caused by rounding, denominator choices, or correlation-matrix construction.

A short illustrative construction is:

```
vy <- yy / n
cov_y <- diag(sqrt(vy)) %*% R_y %*% diag(sqrt(vy))
SSY <- Nmat * cov_y
diag(SSY) <- yy
```

This is a reconstruction under assumptions, not an observed cross-product unless the input covariance and overlap information were obtained directly from the relevant individuals.

11 First-Principles Assumptions

The reconstruction

$$\widehat{SSY}_{ts} = N_{ts}\hat{\sigma}_{ts}$$

rests on several assumptions.

11.1 Centered phenotypes

For nonzero means:

$$E(y_t y_s) = \text{Cov}(y_t, y_s) + E(y_t)E(y_s).$$

The covariance-only reconstruction is therefore appropriate when phenotypes are centered, residualized with an intercept, or otherwise have negligible means on the relevant sample.

11.2 Random or ignorable missingness

The shared individuals $\mathcal{J}_{ts}$ should not have a systematically different cross-trait covariance from the target population. If overlap is selected by disease status, cohort, measurement protocol, or other trait-related factors, covariance homogeneity may fail.

11.3 Homogeneous covariance

The derivation assumes:

$$\text{Cov}(y_{it}, y_{is}) = \sigma_{ts}$$

for individuals contributing to the pairwise product. Cohort heterogeneity, ancestry differences, environmental strata, and phenotype harmonization can violate this assumption.

11.4 Correct overlap counts

The approximation uses the actual or defensible effective value of N_{ts} . Rounding overlap fractions, using total study size instead of shared size, or ignoring duplicated participants changes the implied cross-product.

11.5 Compatible phenotype covariance

The correlation or covariance estimate must refer to phenotypes on compatible scales, with compatible covariate adjustment and trait definitions.

12 Why SSY Matters for Residual Covariance

Let:

$$R_Y = Y - XB$$

be the individual-level residual matrix. Its cross-product is:

$$\begin{aligned} R_Y^\top R_Y &= (Y - XB)^\top (Y - XB) \\ &= Y^\top Y - Y^\top XB - B^\top X^\top Y + B^\top X^\top XB. \end{aligned}$$

Using summary-statistic notation:

$$R_Y^\top R_Y = SSY - WY^\top B - B^\top WY + B^\top X^\top XB.$$

The off-diagonal entries of this residual cross-product depend directly on the off-diagonal entries of **SSY**. Without them, a sampler cannot reconstruct the full residual cross-product from marginal summaries alone.

The current CSR MTBLR implementation accepts a full $T \times T$ matrix through its **yy/YY** argument. It combines that matrix with marker-trait cross-products, current effects, and marker-scale residual scores, then symmetrizes the resulting residual cross-product before updating the residual covariance matrix.

If only diagonal **yy** values are available, a model can still update trait-specific residual variances. A full residual covariance update then requires off-diagonal values to be fixed, reconstructed, externally supplied, or excluded.

13 Genetic Covariance and Residual Covariance

Multi-trait models can borrow information through both marker-effect and residual covariance, but these sources must not be conflated.

13.1 Marker-effect covariance

For marker j , let:

$$b_j = [b_{j1} \quad \cdots \quad b_{jT}]^\top.$$

A multi-trait prior can assign:

$$b_j \mid \text{active model} \sim N_T(0, \Sigma_b)$$

on the active traits. Off-diagonal entries of Σ_b allow effects to co-vary across traits. Trait-sharing model indicators can additionally encode whether a marker is active for specific subsets of traits.

13.2 Genetic covariance from fitted effects

With a common genotype matrix:

$$G_X = XB$$

has cross-product:

$$G_X^\top G_X = B^\top X X B.$$

Using marker-score residuals $R = WY - X X B$, a symmetric summary-statistic estimate for traits t and s is:

$$\widehat{G}_{ts} = \frac{b_t^\top (wy_s - r_s) + b_s^\top (wy_t - r_t)}{2\sqrt{n_t n_s}}.$$

This quantity concerns genetic values induced by marker effects and LD. It is not determined by SSY alone.

13.3 Residual covariance

Residual covariance concerns phenotype variation not represented by the current marker effects. Its observed cross-trait information comes from shared individuals and therefore depends on SSY and sample overlap. A genetic correlation may exist with no sample overlap; direct residual covariance information cannot.

14 Connection to Current `sblr` Interfaces

The public `sblr()` wrapper describes `yy` as a trait cross-product matrix, `Xy` as marker-trait cross-products, and `XX` as marker cross-product or LD matrices. It initializes trait variances from the diagonal of `yy`. However, the native contracts differ across its legacy algorithm choices: several older generated interfaces expose `yy` as a vector, whereas the CSR MTBLR kernel described below explicitly consumes a full matrix. Users and developers must verify the exact selected algorithm before assuming that off-diagonal SSY entries are used.

The generated `mtblr_cpg_omp_csr()` interface accepts:

- `wy`, one marker vector per trait;
- `ww`, marker diagonal cross-products per trait;
- `yy`, a full $T \times T$ trait cross-product matrix;
- `ld_prefix`, identifying shared CSR LD;
- marker-effect covariance `B`;
- residual covariance `E`;
- multi-trait sharing models and model probabilities; and
- a sample-size vector `n`.

The current CSR kernel requires equal values in `n` and matching `ww` across traits because it assumes shared LD scaling. It does not accept `Nmat` directly. Therefore, `Nmat` currently serves as an analysis-level object used to construct, justify, and validate `SSY`; it is not automatically used by the native CSR likelihood to adjust each trait pair.

This distinction matters. A mathematically reconstructed `SSY` for partial overlap does not by itself remove the kernel's shared- X , shared-LD, and equal- n assumptions. More general partial-overlap likelihoods require additional implementation and validation.

15 Simulation as a Known-Truth Check

`mtsim()` and `mtsim_annotation()` simulate all traits on one common genotype matrix and one common individual set. They generate:

- marker-effect matrix `B`;
- genetic values `G`;
- residual matrix `E`;
- phenotype matrix `y`;
- target and observed genetic correlations; and
- target and observed residual correlations.

For such simulations, full-overlap quantities can be calculated directly:

```
Y <- as.matrix(sim$y)
SSY <- crossprod(Y)
Nmat <- matrix(nrow(Y), nrow = ncol(Y), ncol = ncol(Y))
```

These simulations are useful for checking whether an MTBLR implementation recovers known covariance and sharing patterns. They do not validate partial-overlap reconstruction assumptions unless missingness or study separation is explicitly simulated.

The existing workflow scripts currently use the simulated phenotype columns primarily in parallel single-trait CSR and annotation-aware fits. A complete joint MTBLR workflow must additionally construct and validate full cross-trait inputs.

16 Practical Data Requirements

A joint summary-statistics MTBLR analysis should retain:

Object	Required contract
Trait names/order	Identical across every trait-indexed object
$WY = X^T Y$	$m \times T$, with marker and trait order recorded
<code>ww</code> / diagonal XX	Length m per trait; compatible with LD scaling
LD / XX	Marker order identical to rows of WY
<code>SSY</code>	Symmetric $T \times T$ observed or reconstructed trait cross-product matrix

Object	Required contract
n	Trait-specific sample-size vector, even when all values are equal
Nmat	Symmetric $T \times T$ pairwise overlap matrix used to justify SSY
Phenotype scaling	Centering, variance scaling, transformations, and covariate adjustment
Covariance priors	Trait order and units matching SSY and marker statistics

For current CSR MTBLR use, additionally verify that the shared-LD and equal sample-size restrictions are satisfied.

17 Validation Checks

Before fitting or interpreting a joint multi-trait model:

1. Confirm that **SSY** is $T \times T$, finite, and symmetric.
2. Confirm that **diag(SSY)** equals the trait-specific **yy** values.
3. Check that observed full-overlap **SSY** = **crossprod(Y)** is positive semidefinite up to numerical tolerance.
4. For reconstructed pairwise **SSY**, inspect eigenvalues and understand that pairwise-valid entries do not automatically guarantee a globally positive semidefinite matrix.
5. Confirm that **Nmat** is symmetric and finite.
6. Confirm that:

$$N_{tt} = n_t$$

and:

$$0 \leq N_{ts} \leq \min(n_t, n_s).$$

7. Confirm that trait order is identical across **SSY**, **Nmat**, **WY**, n , Σ_b , Σ_e , prior scale matrices, and outputs.
8. Confirm that **WY**, **ww**, and **LD** use the same marker order and compatible genotype scaling.
9. Confirm that reported phenotype correlations are on compatible phenotype definitions and scales.
10. Record whether every off-diagonal **SSY** value was observed, reconstructed, fixed, or assumed.
11. For current CSR MTBLR, confirm equal **n** values and equal **ww** vectors across traits.

12. Inspect covariance traces, eigenvalues, correlations, and sensitivity to overlap assumptions.

Positive semidefiniteness is especially important because covariance updates require valid scale matrices. Small negative eigenvalues may arise from rounding and can sometimes be regularized; substantial negative eigenvalues usually indicate incompatible pairwise inputs or assumptions.

18 Limitations and Sensitivity Analysis

Multi-trait summary-statistics inference is sensitive to assumptions that are unnecessary or directly observable with complete individual-level data.

- **Overlap uncertainty:** errors in N_{ts} directly alter reconstructed **SSY** off-diagonals.
- **Nonrandom missingness:** shared individuals may not represent the marginal trait samples.
- **Phenotype heterogeneity:** correlations can differ across cohorts, ancestries, environments, or phenotype definitions.
- **LD mismatch:** trait-specific association statistics may not be compatible with one shared LD reference.
- **Scale mismatch:** correlations, yy , WY , and covariance priors may be expressed on different phenotype scales.
- **Weak identification:** residual covariance is weakly or not directly identified when overlap is small, zero, or unknown.
- **Covariance instability:** high-dimensional or highly correlated trait covariance matrices can be numerically unstable.
- **Current implementation scope:** the CSR MTBLR kernel's shared-LD and equal- n requirements are narrower than the most general partial-overlap theory.

Production analyses should compare plausible overlap matrices, phenotype correlation sources, residual covariance restrictions, and LD references. Repeated chains, covariance diagnostics, predictive validation, and sensitivity to regularization are required before interpreting cross-trait borrowing or covariance estimates.

19 Related Notes and Sources

- [Technical summary statistics](#) derives the general cross-product and residual-score representation.
- [Multi-trait models](#) provides the conceptual model map for covariance and trait-sharing structures.
- [Data representations](#) introduces individual-level and summary-statistics inputs.
- [Model overview](#) places MTBLR within the broader **sblr** workflow landscape.

This appendix consolidates every mathematical component of the archived **SSY** and sample-overlap theory while aligning its claims with the current multi-trait wrappers and CSR implementation.